



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS & COMPUTER SCIENCE

End Semester Examination – 2023/2024 (Semester II)
STA1023/MAT1063/ST102/MT102 – Introduction to Probability Theory

Time allowed: **TWO** hours

Answer **ALL** questions

Non-programmable, scientific calculators are allowed.

Statistical tables are provided

- 1) Counting techniques are useful to calculate all possible ways of completing a task under different conditions.
 - a) Suppose a job consists of k different tasks where the i^{th} of which can be done in n_i ways, $i = 1, \dots, k$.
Show that the entire job can be done in $n_1 \times n_2 \times \dots \times n_k$ different ways. [30 marks]
 - b) Four boys and four girls are to be seated at a round table.
 - i. How many distinct seating arrangements are there if no restrictions are imposed? [5 marks]
 - ii. In how many distinct ways can 4 boys and 4 girls be seated alternately? [10 marks]
 - iii. In how many ways can they be seated if one boy and one girl among them refused to sit together? [10 marks]
 - iv. Derive a general formula for part (iii) if a total of m boys and n girls appear in the scenario where one boy and one girl among them refused to sit together. [15 marks]
 - c) Morse code is used to send messages with “dots” and “dashes”, which are known to occur in a proportion of 3:4. Suppose there is an interference in the transmission line, and with probability $1/8$ a dot is mistakenly received as a dash and vice versa. Calculate the probability of receiving a dot correctly. [30 marks]
- 2) Two fair six-sided dice are rolled simultaneously. Define a random variable X such that;
 $X =$ the sum of the outcomes of the two dice.
 - a) Derive the probability mass function (pmf) of X . [20 marks]
 - b) Verify that function derived in part (a) is a legitimate pmf. [10 marks]
 - c) Calculate $P(X \text{ is an even number})$. [10 marks]
 - d) Calculate the expected value and the variance of X . [20 marks]
 - e) Derive the Cumulative Density Function (CDF) of X and sketch the function. [20 marks]
 - f) Explain how the CDF can be used to find the median of X . [20 marks]
- 3) a) Let X be a Poisson random variable with parameter λ .
 - i. Show that $E[X^n] = \lambda E[(X+1)^{n-1}]$ for $n \geq 1$. [35 marks]
 - ii. Using the result in part (i), compute $E[X^3]$. [15 marks]
- b) A loss for a company has the moment-generating function $M(t) = \frac{0.16}{0.16-t}$, where $t < 0.16$. An insurance policy pays a benefit equal to 70% of the loss. What is the moment-generating function of the benefit? [20 marks]
[P.T.O]

[Question 3 continued...]

- c) A lot of 40 components is deemed unacceptable if it contains 3 or more defectives. The procedure for sampling a lot is to select 5 components at random and to reject the lot if at least one defective is found. What is the probability that exactly 1 defective is found in the sample, given that there are exactly 3 defectives in the entire lot?

[30 marks]

4)

- a) Buses arrive at a specified stop at 15-minute intervals starting at 7:00 A.M. That is, they arrive at 7:00, 7:15, 7:30, 7:45, and so on. The arrival time of a passenger at this stop is uniformly distributed between 7:00 and 7:30. Find the probability that the passenger waits;

i. less than 5 minutes for a bus;

[25 marks]

ii. more than 10 minutes for a bus.

[15 marks]

- b) The company “Bright Lights” manufactures a new line of LED bulbs. The quality control department has determined that the lifespan (in hundred hours) of these bulbs, X , follows a normal distribution, $X \sim N(\mu, \sigma^2)$.

If 12% of bulbs fail before 72 (hundred hours) and the top 7% last longer than 109 (hundred hours), find the values of the mean μ and the standard deviation σ .

[60 marks]

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Distribution	Density Function
Binomial (n, θ)	$P(X = x) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}; x = 0, 1, 2, \dots, n$
Negative Binomial (k, θ)	$P(X = x) = \binom{x-1}{k-1} \theta^k (1 - \theta)^{x-k}; x = k, k+1, k+2, \dots$
Geometric (θ)	$P(X = x) = \theta(1 - \theta)^{x-1}; x = 1, 2, \dots$
Hypergeometric (n, N, M)	$P(X = x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}; x = 0, 1, 2, \dots, n, x \leq M \text{ and } n - x \leq N - M$
Poisson (λ)	$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}; x = 0, 1, 2, \dots$